Università degli studi di Padova

Stochastic Methods for Finance

Report 2: Implied Term Structure and Dividends from Equity Option Prices

Proff.
Martino Grasselli
Giorgia Callegaro

Student
Angela Bortolato
2156562

Academic Year 2025-2026

1 Company Profile and Dividend Policy

Microsoft Corporation (MSFT) is a global leader in software and cloud computing (Azure, Microsoft 365). With a market capitalization of approx. \$2.84T and robust annual net income (\$119.26B), the company maintains a high-quality credit profile. For this analysis, MSFT is an ideal candidate due to its long history of consistent, growing quarterly dividends (Table 1).

Date	**Div (USD)**	**Date**	**Div (USD)**
Feb 19, 2026	0.91	Aug 21, 2025	0.83
Nov 20, 2025	0.91	May 15, 2025	0.83

Table 1: Recent MSFT Dividend Payments.

2 Option Data and Discounting

The dataset in Table 2 comprises bid-ask quotes for MSFT options (retrieved from Refinitiv, March 27, 2026) with an underlying price $S_0 = 359.22$. Although legally American, early exercise premia are assumed negligible.

The implied discount factor $D(0,T)$ is estimated via the no-arbitrage relation:

$$D(0,T) = \frac{C(T,K_1) - C(T,K_2) + P(T,K_2) - P(T,K_1)}{K_2 - K_1}$$

To ensure robustness against liquidity noise and bid-ask spreads, $D(0,T)$ is calculated across multiple strike pairs (K_1, K_2) centered around S_0. The continuously compounded rate is then $r = -\ln D(0,T)/T$. Results are shown in Table 3.

3 Implied Dividends

From put-call parity at a given strike K around the money
$C(T,K) - P(T,K) = D(0,T) \cdot (F(0,T) - K)$ we first extract the forward price

$$F(0,T) = \frac{C(T,K) - P(T,K)}{D(0,T)} + K$$

Using the relation $S_0 - (F(0,T) + \text{div}_T) \cdot D(0,T) = 0$ we obtain the implied dividends as

$$\text{div}_T = \frac{S_0}{D(0,T)} - F(0,T)$$

where div_T represents the future value of dividends accumulated over the time interval $[0,T]$.

The implied dividends are computed for strikes close to the current underlying price S_0, in order to reduce the impact of bid-ask spreads and liquidity effects. The final estimate is

Strike price K	Call bid	Call ask	Call mid	Put bid	Put ask	Put mid
Option expiry 24/04/2026 - Maturity 1 month						
335	23.65	30.45	27.05	4.8	5.1	4.95
340	25.65	26.5	26.08	5.75	6.15	5.95
345	21.35	22.75	22.05	7	7.4	7.20
350	18.5	19.35	18.93	8.5	8.85	8.68
355	15.4	16.35	15.88	10.25	10.65	10.45
360	12.5	13.1	12.80	12.35	12.7	12.53
365	10.1	10.45	10.28	14.55	15.2	14.88
370	7.7	8.35	8.03	17.55	18.05	17.80
375	6.25	6.45	6.35	20.6	21.35	20.98
380	4.45	4.95	4.70	24.1	24.95	24.53
Option expiry 18/06/2026 - Maturity 3 months						
335	37.75	40.15	38.95	13.15	13.3	13.23
340	34.75	35.7	35.23	14.7	14.9	14.80
345	31.2	33.4	32.30	16.45	16.65	16.55
350	29.7	30.3	30.00	18.35	18.55	18.45
355	27	27.2	27.10	20.4	20.6	20.50
360	24.2	24.45	24.33	22.6	22.85	22.73
365	21.65	21.85	21.75	25.1	25.3	25.20
370	19.3	19.5	19.40	27.65	27.85	27.75
375	17.05	17.25	17.15	30.4	30.7	30.55
380	15.05	15.25	15.15	33.4	33.65	33.53
Option expiry 18/09/2026 - Maturity 6 months						
310	65.65	68.3	66.98	13.8	13.95	13.88
320	59.1	61	60.05	16.5	16.65	16.58
330	51.65	54.15	52.90	19.55	19.7	19.63
340	45.3	47.65	46.48	23.05	23.2	23.13
350	39.9	41.55	40.73	27	27.2	27.10
360	35.55	35.85	35.70	31.45	31.65	31.55
370	30.6	30.85	30.73	36.3	36.55	36.43
380	26.1	26.35	26.23	41.8	42.05	41.93
390	22.1	22.3	22.20	47.75	48.1	47.93
400	18.55	18.8	18.68	53.85	55.2	54.53
Option expiry 19/03/2027 - Maturity 1 year						
310	79.25	81.5	80.38	22.2	22.6	22.40
320	72	73.25	72.63	25.45	25.9	25.68
330	65.5	66.2	65.85	29	29.4	29.20
340	61	62.7	61.85	32.95	33.35	33.15
350	55.55	57.2	56.38	37.2	37.6	37.40
360	51.35	52.25	51.80	41.8	42.25	42.03
370	46.65	47.1	46.88	46.8	47.2	47.00
380	42.1	42.5	42.30	52.15	52.55	52.35
390	37.9	38.35	38.13	57.85	58.35	58.10
400	34.05	34.4	34.23	63.95	64.45	64.20

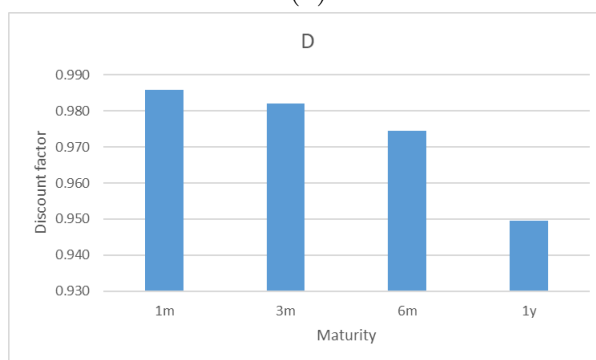
Table 2: Option prices for MSFT across different strikes and maturities, retrieved from Refinitiv on March 27, 2026.

K_1	K_2	$D(0,T)$	r
Maturity 1 month			
355	360	1.030	-0.355
350	365	0.990	0.121
345	370	0.985	0.181
340	375	0.993	0.086
335	380	0.932	0.849
Maturity 3 months			
355	360	1.000	0.000
350	365	1.000	0.000
345	370	0.964	0.147
340	375	0.966	0.137
335	380	0.980	0.081
Maturity 6 months			
350	360	0.947	0.108
340	370	0.968	0.064
330	380	0.980	0.041
320	390	0.989	0.023
310	400	0.988	0.023
Maturity 1 year			
350	360	0.920	0.083
340	370	0.961	0.040
330	380	0.934	0.068
320	390	0.956	0.045
310	400	0.977	0.023

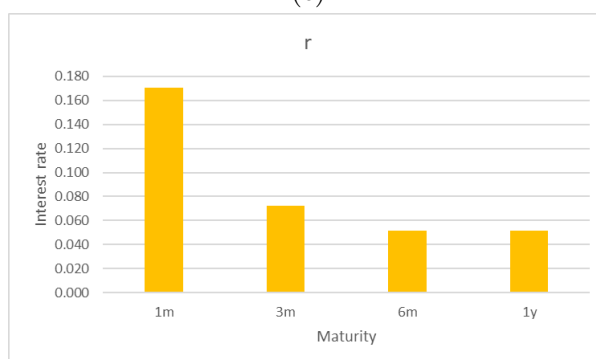
(a)

Maturity T	$D(0,T)$	r
1 month	0.986	0.170
3 months	0.982	0.072
6 months	0.974	0.052
1 year	0.950	0.052

(b)



(c)



(d)

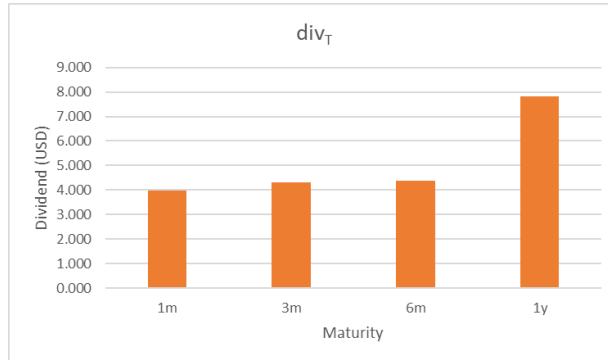
Table 3: Estimation of the discount factor and implied interest rate across maturities. (a) Discount factors $D(0,T)$ and corresponding interest rates r computed for different pairs of strikes (K_1, K_2); (b) aggregated estimates obtained by averaging across strike pairs; (c) term structure of the discount factor $D(0,T)$ as a function of maturity; (d) term structure of the implied interest rate r.

K	F	div_T(USD)	q
Maturity 1 month			
355	360.503	3.853	0.128
360	360.279	4.077	0.135
365	360.334	4.021	0.133
Maturity 3 months			
355	361.720	4.205	0.046
360	361.629	4.296	0.047
365	361.487	4.438	0.049
Maturity 6 months			
350	363.982	4.534	0.025
360	364.259	4.258	0.023
370	364.151	4.366	0.024
Maturity 1 year			
350	369.982	7.989	0.021
360	370.294	7.677	0.021
370	369.868	8.102	0.022

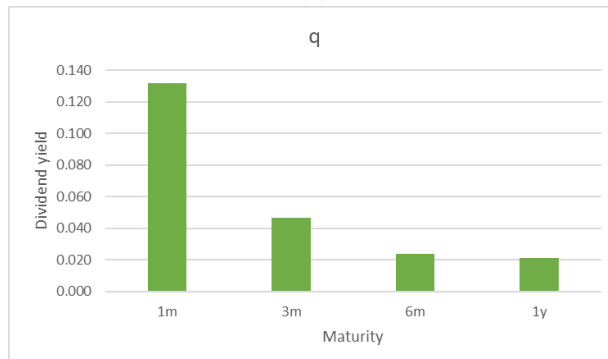
(a)

Maturity T	div_T	q
1 month	3.984	0.132
3 months	4.313	0.047
6 months	4.386	0.024
1 year	7.922	0.021

(b)



(c)



(d)

Table 4: Implied dividends and dividend yield across maturities: (a) detailed results by strike, (b) aggregated estimates, (c) term structure of implied dividends, (d) term structure of the dividend yield.

obtained by averaging the values corresponding to the selected strikes and shown in Table 4.

Although the options considered are American-style, we assume that early exercise effects are negligible for the purposes of this analysis.

Assuming a continuous dividend yield, the forward price satisfies $F(0,T) = S_0 e^{(r-q)T}$, where q denotes the dividend yield.

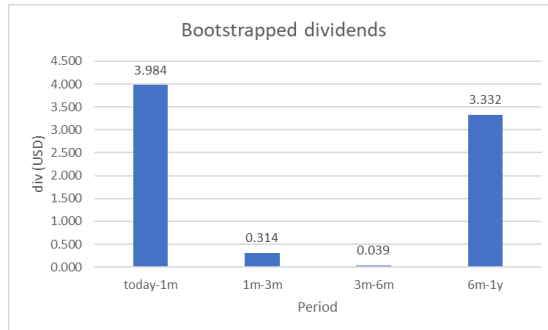
Using the previously estimated forward price and interest rate, the implied dividend yield can be recovered as

$$q = r - \frac{1}{T} \ln \left(\frac{F(0,T)}{S_0} \right).$$

The yield is computed for strikes close to the current underlying price and averaged to obtain a stable estimate.

T	$PV(div)$	period	$PV(D_i)$	D_i
1m	3.928	today-1m	3.928	3.984
3m	4.236	1-3m	0.308	0.314
6m	4.274	3-6m	0.038	0.039
1y	7.438	6m-1y	3.164	3.332

(a)



(b)

Table 5: Bootstrapping results

3.1 Bootstrapping the Dividend Term Structure

The implied dividends obtained for each maturity represent cumulative cash flows over the interval $[0, T]$. In order to extract the term structure of dividends, a bootstrapping procedure is applied.

First, cumulative dividends are converted into present value terms $PV(\text{div}_T) = D(0, T)\,\text{div}_T$. Then, present value individual dividends over successive intervals are obtained recursively as

$$PV(D_i) = PV(\text{div}_{T_i}) - PV(\text{div}_{T_{i-1}})$$

Finally, to obtain forward schedule of dividends it is needed to re-capitalize.
This procedure allows the extraction of the dividend term structure from option prices, results are provided in Table 5.

4 Discussion of Results

The empirical results align qualitatively with no-arbitrage theory, showing a decreasing discount factor and an increasing cumulative dividend schedule. The implied interest rate curve exhibits a downward slope, likely reflecting market expectations or liquidity premiums in longer-dated tenors.

The dividend term structure successfully captures the discrete nature of Microsoft's payouts. The observed fluctuations in the implied yield and the slight overestimation of dividends highlight the sensitivity of the model to the input and the limitations of applying European put-call parity to American-style options without adjusting for early exercise premia.